Assignment 2: Topic Modeling Analysis of Airbnb Reviews

Objective:

The objective of this assignment is to compare the hidden themes in Airbnb reviews from Asheville during the pre- and post-pandemic periods. For the purposes of this analysis, 2019 will represent the pre-pandemic year and 2023 will represent the post-pandemic period.

Use the file titled "W7_Topic Modeling: Discovering Hidden Themes in Airbnb Reviews" (from the Week 7 module) as your base notebook. This file contains the complete code for analyzing the 2019 (pre-pandemic) Airbnb reviews using a sample of 10,000 reviews. You may conduct the analysis on the full 2019 dataset or randomly select a sample if computational time becomes excessive, as demonstrated in the provided code.

Afterward, apply the same analytical procedures to the 2023 (post-pandemic) dataset using a comparable sample size. The assignment involves determining the optimal number of topics, labeling these topics appropriately, and deriving business insights relevant to Airbnb stakeholders. You may also experiment with BERTopic, although its use is optional due to its computational intensity.

Assignment Instruction:

Base file: *W7_Topic Modeling: Discovering Hidden Themes in Airbnb Review:* We used this notebook in class for practice, which includes the full topic modeling workflow for 2019 (pre-pandemic) data for a random sample of 10,000 reviews. It covers data preparation and preprocessing steps, implementations of LDA, NMF, LSA, and BERTopic models, as well as code for visualizing the 2019 topics.

Data Sources: Asheville Reviews

Your Task:

1. Determine the appropriate number of topics for each model (LDA, NMF, and LSA) and run the analysis in the provided notebook for 2019 data. You may update the custom stopwords list to improve the relevance of the topics. Be sure to carefully review the code and comments in the provided notebook.
2. Modify the notebook to analyze 2023 (post-pandemic) data and re-run the models.
3. Label the topics based on the results for both pre- and post-pandemic periods.
4. Write a business report highlighting the findings and implications as discussed below
5. ***Optional:*** If you have the resources, explore BERTopic to compare its results with the other models.

Grading Scale:

Part 1: Conducting Topic Modeling (10 points)

Part 2: Business Report (20 points)

This is the most important part of the assignment, where the majority of the points are allocated.

1. Topic Labeling:
    - Provide meaningful labels for the topics in both pre- and post-pandemic data. Interpret the significance of these topics.
2. Discuss and Compare the Results:
    - Discuss your results in detail. Compare the topics generated for 2019 and 2023 across all models (LDA, NMF, LSA).
    - Discuss whether there are any changes in the dominant topics between the pre-pandemic and post-pandemic periods.
3. Stakeholder Insights and Recommendations:
    - Based on your analysis, what recommendations can you provide to Airbnb management? What topics seem most important to customers?
    - What other types of analysis would you recommend providing further insights?

Deliverables:

1. **Business Report (in Word or pdf format) (20 points):**
    - **Note: Adding comments to the Jupyter notebook does not constitute a business report. You must submit a formal report**.
    - The report should summarize your findings and include relevant plots and labeled topics. Discuss the findings, insights, and business recommendations for Airbnb stakeholders as mentioned above.
2. **Completed Python Notebook (10 points):**
    - Submit the modified Python notebook where you analyzed the 2023 data using the provided models.

Criteria	Excellent (Full Points)	Satisfactory (Partial Points)	Needs Improvement (Few or No Points)	Points
1. Conducting Topic Modeling (10 points) – Jupyter Notebook				

Data Preparation & Preprocessing (4 pts)	Properly loads and processes 2023 data, ensures consistency with 2019 dataset, updates stopword list, and documents preprocessing steps. (4 pts)	Loads and processes 2023 data but lacks consistency with 2019 dataset or has incomplete documentation. (2-3 pts)	Fails to correctly process 2023 data, missing major preprocessing steps or documentation. (0-1 pts)	**4**
Running Topic Models (6 pts)	Successfully implements LDA, NMF, and LSA on 2023 data, determining optimal topic numbers and justifying choices. (6 pts)	Runs models but lacks justification for topic selection or has minor errors. (3-5 pts)	Major errors in running models, missing models, or incorrect implementation. (0-2 pts)	**6**
2. Business Report (20 points)				
Topic Labeling (4 pts)	Provides clear, meaningful, and well-justified labels for all topics in both 2019 and 2023 datasets. (4 pts)	Labels are somewhat meaningful but lack depth or justification. (2-3 pts)	Topic labels are unclear, generic, or missing. (0-1 pts)	**4**
Comparison of Pre- and Post-Pandemic Topics (6 pts)	Thoughtful comparison of topics, highlighting key differences and trends across models. (6 pts)	Compares topics but lacks depth or only focuses on one model. (3-5 pts)	Minimal or no comparison between 2019 and 2023 topics. (0-2 pts)	**6**
Stakeholder Insights & Recommendations (6 pts)	Provides meaningful insights for Airbnb, hosts, and guests, with actionable recommendations. (6 pts)	Some insights provided but lacks strong recommendations. (3-5 pts)	Minimal or no insights relevant to stakeholders. (0-2 pts)	**6**
Report Structure & Clarity (4 pts)	Well-structured, professional report with clear explanations, relevant plots, and strong conclusions. (4 pts)	Report is mostly clear but lacks organization or some necessary visualizations. (2-3 pts)	Poorly structured report, missing key elements or explanations. (0-1 pts)	**4**
				30